

Foundation (Jalapeno)

Q1. Rationalize the denominator: $\frac{1}{\sqrt{2}}$

- (A) $\frac{1}{2}$
- (B) $\frac{\sqrt{2}}{2}$
- (C) $\frac{2}{\sqrt{2}}$
- (D) $\frac{\sqrt{2}}{\sqrt{2}}$
- (E) $\frac{2}{2\sqrt{2}}$

Q2. Rationalize the denominator: $\frac{1}{\sqrt{3}}$

- (A) $\frac{\sqrt{3}}{3}$
- (B) $\frac{1}{3}$
- (C) $\frac{3}{\sqrt{3}}$
- (D) $\frac{\sqrt{3}}{\sqrt{3}}$
- (E) $\sqrt{3}$

Q3. Rationalize the denominator: $\frac{2}{\sqrt{5}}$

- (A) $\frac{2\sqrt{5}}{5}$
- (B) $\frac{\sqrt{5}}{5}$
- (C) $\frac{5}{2\sqrt{5}}$
- (D) $\frac{2}{5}$
- (E) $\frac{\sqrt{5}}{2}$

Q4. Rationalize the denominator: $\frac{3}{\sqrt{2}}$

- (A) $\frac{3\sqrt{2}}{2}$
- (B) $\frac{3}{2}$
- (C) $\frac{\sqrt{2}}{3}$
- (D) $\frac{2}{3\sqrt{2}}$
- (E) $\frac{\sqrt{2}}{2}$

Q5. Rationalize the denominator: $\frac{\sqrt{2}}{\sqrt{3}}$

- (A) $\frac{\sqrt{6}}{3}$
- (B) $\frac{\sqrt{2}}{3}$
- (C) $\frac{\sqrt{3}}{\sqrt{2}}$
- (D) $\frac{2}{\sqrt{3}}$
- (E) $\frac{\sqrt{2}\sqrt{3}}{3}$

Q6. Rationalize the denominator: $\frac{2\sqrt{3}}{\sqrt{2}}$

- (A) $\frac{2\sqrt{6}}{2}$
- (B) $\frac{\sqrt{3}}{\sqrt{2}}$
- (C) $\frac{2\sqrt{3}}{2}$
- (D) $\sqrt{6}$
- (E) $\frac{2\sqrt{2}\sqrt{3}}{2}$

Q7. Rationalize the denominator: $\frac{1}{\sqrt{6}}$

- (A) $\frac{\sqrt{6}}{6}$
- (B) $\frac{\sqrt{2}\sqrt{3}}{6}$
- (C) $\frac{1}{6}$
- (D) $\frac{\sqrt{2}}{\sqrt{3}}$
- (E) $\frac{\sqrt{3}}{\sqrt{2}}$

Q8. Rationalize the denominator: $\frac{\sqrt{2}}{\sqrt{8}}$

- (A) $\frac{\sqrt{2}}{4}$
- (B) $\frac{\sqrt{2}}{2\sqrt{2}}$
- (C) $\frac{1}{2}$
- (D) $\frac{\sqrt{2}}{2\sqrt{2}}$
- (E) $\frac{1}{\sqrt{2}}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator and simplify: $\frac{2+\sqrt{3}}{\sqrt{2}-\sqrt{3}}$

Q10. Rationalize the denominator and simplify: $\frac{\sqrt{5}+1}{\sqrt{2}+\sqrt{5}}$

Chef's Tips

- Check for radical simplification before rationalizing the denominator.
- Make sure to multiply by the conjugate when rationalizing the denominator.
- Be mindful of sign changes when simplifying rationalized expressions.